

Time-development of the concrete secant modulus of elasticity, from cement class and mean 28-day strength. Shared upstream of crack-width, deflection and bending-stiffness calculations.

1. Input Parameters

Symbol	Value	Unit	Description
S_{cem}	0.25	-	Cement class coefficient (0.20 R / 0.25 N / 0.38 S)
t	7	days	Concrete age at loading
f_{cm}	38	MPa	Mean compressive strength at 28 days (C30/37)
E_{cm}	32837	MPa	Secant modulus at 28 days (C30/37)

2. Calculation (all formulas from the shared library)

Concrete strength development coefficient at age t [EN 1992-1-1:2004 §3.1.2(6), Eq. (3.2)]

$$\beta_{cc,t} = e^{S_{cem} \cdot \left(1 - \sqrt{\frac{28}{t}}\right)}$$
$$= e^{0.25 \cdot \left(1 - \sqrt{\frac{28}{7}}\right)}$$

= **0.7788**

Mean concrete compressive strength at age t [EN 1992-1-1:2004 §3.1.2(6), Eq. (3.1)]

$$f_{cm,t} = \beta_{cc,t} \cdot f_{cm}$$
$$= 0.7788 \cdot 38$$

= **29.59 MPa**

Secant modulus of elasticity at age t [EN 1992-1-1:2004 §3.1.3(3), Eq. (3.5)]

$$E_{cm,t} = E_{cm} \cdot \left(\frac{f_{cm,t}}{f_{cm}}\right)^{0.3}$$
$$= 32837 \cdot \left(\frac{29.59}{38}\right)^{0.3}$$

= **30464 MPa**

3. Verification

OK

$E_{cm,t} = 30464 \text{ MPa} \leq E_{cm} = 32837 \text{ MPa}$ — Modulus at early age does not exceed the 28-day value